Piecewise-Defined & Implicit Functions

ANSWER KEY

Section 1: Evaluating Piecewise Functions

1 Let $f(x) = 2x + 1$ for $x < 2$, and $f(x) = x^2 - 3$ for $x \geq 2$. Find $f(2)$ and $f(-1)$.

a. $f(2) = 1$ **and** $f(-1) = -1$ ✓

b. $f(2) = 5$ and $f(-1) = -1$

c. $f(2) = 1$ and $f(-1) = -2$

d. $f(2) = 5$ and $f(-1) = 1$

For $x = 2$, use the second piece: $f(2) = 2^2 - 3 = 1$. For $x = -1 < 2$, use the first piece: $f(-1) = 2(-1) + 1 = -1$.

2 A tax bracket levies no tax on the first 10,000 dollars, and 15% on income exceeding 10,000 dollars. Which piecewise function $T(x)$ models total tax for income $x \geq 0$?

a. $T(x) = 0$ **for** $x \leq 10000$**, and** $T(x) = 0.15(x - 10000)$ **for** $x > 10000$ ✓

b. $T(x) = 0.15x$ for all x

c. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15x$ for $x > 10000$

d. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 1500 + 0.15x$ for $x > 10000$

No tax is paid on income up to 10,000 ($T = 0$). Only the amount beyond 10,000 ($(x - 10000)$) is taxed at 15%, yielding $T(x) = 0.15(x - 10000)$.

3 Evaluate the ceiling function $g(x) = \lceil 2x - 0.3 \rceil$ at $x = 1.4$.

a. 3 ✓

b. 2

c. 2.5

d. 4

$2(1.4) - 0.3 = 2.8 - 0.3 = 2.5$. The ceiling function rounds up to the least integer greater than or equal to 2.5, which is 3.

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Section 2: Continuity of Piecewise Functions at Boundaries

4 Find the value of k that makes $f(x)$ continuous at $x = 3$, where $f(x) = kx + 1$ for $x \leq 3$ and $f(x) = 2x^2 - 5$ for $x > 3$.

a. 4 ✓

b. 2

c. 5

d. 13

For continuity at $x = 3$, $\lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x)$. Left limit: $k(3) + 1 = 3k + 1$. Right limit: $2(3^2) - 5 = 18 - 5 = 13$. Setting $3k + 1 = 13 \Rightarrow 3k = 12 \Rightarrow k = 4$.

5 Let $g(x) = \frac{x^2 - 16}{x - 4}$ for $x \neq 4$, and $g(4) = c$. What value of c removes the discontinuity at $x = 4$?

a. 8 ✓

b. 4

c. 0

d. 16

$\lim_{x \to 4} \frac{(x-4)(x+4)}{x-4} = \lim_{x \to 4} (x + 4) = 8$. For g to be continuous at $x = 4$, c must equal the limit, so $c = 8$.

6 A function h is defined by $h(x) = |x - 2| + 3$. Where is $h(x)$ continuous and where is it not differentiable?

a. **Continuous everywhere; not differentiable at** $x = 2$ ✓

b. Continuous only for $x \geq 2$; differentiable everywhere

c. Discontinuous at $x = 2$; differentiable elsewhere

d. Continuous and differentiable everywhere on $(-\infty, \infty)$

Absolute value functions are continuous everywhere. However, at $x = 2$ there is a sharp corner, where the left-hand slope (-1) differs from the right-hand slope $(+1)$, so it is not differentiable at $x = 2$.

Section 3: Implicit Relations and Conic Sections

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7 Identify the conic section described by the equation $4x^2 + 9y^2 - 16x + 54y + 61 = 0$.

a. **Ellipse** ✓

b. Hyperbola

c. Parabola

d. Circle

Both x^2 and y^2 have positive coefficients ($A = 4, C = 9$) and $A \neq C$. Completing the square confirms this is an ellipse.

8 Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y = 0$.

a. **Center** $(3, -4)$**, Radius** 5 ✓

b. Center $(-3, 4)$, Radius 5

c. Center $(3, -4)$, Radius 25

d. Center $(6, -8)$, Radius 10

Complete the square: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 \Rightarrow (x - 3)^2 + (y + 4)^2 = 25$. Center is $(3, -4)$ and radius is $\sqrt{25} = 5$.

9 Which implicit relation fails the Vertical Line Test and therefore cannot be represented by a single function $y = f(x)$?

a. $x^2 + y^2 = 25$ ✓

b. $y = x^2 - 4$

c. $y = \sqrt{25 - x^2}$

d. $y = 3x + 1$

$x^2 + y^2 = 25$ is a circle centered at $(0, 0)$. For any $x \in (-5, 5)$, there are two corresponding y values ($y = \pm\sqrt{25 - x^2}$), failing the vertical line test.

Section 4: Asymptotes and Branches of Hyperbolas

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10 Find the equations of the asymptotes for the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{3}{4}x$ ✓

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

For a horizontal hyperbola centered at origin $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x$. Here $a = 4, b = 3$, giving $y = \pm\frac{3}{4}x$.

11 What is the orientation of the transverse axis for the hyperbola $\frac{(y-1)^2}{25} - \frac{(x+2)^2}{16} = 1$?

a. **Vertical, along the line** $x = -2$ ✓

b. Horizontal, along the line $y = 1$

c. Diagonal with slope $5/4$

d. Vertical, along the y-axis

Because the positive term contains $(y-1)^2$, the hyperbola opens vertically, and its transverse axis is vertical along the line $x = -2$.

12 What are the vertices of the ellipse $\frac{x^2}{49} + \frac{y^2}{25} = 1$?

a. $(\pm 7, 0)$ ✓

b. $(0, \pm 5)$

c. $(\pm 5, 0)$

d. $(0, \pm 7)$

Since $a^2 = 49 > b^2 = 25$, the major axis is horizontal with $a = 7$. Vertices are at $(\pm a, 0) = (\pm 7, 0)$.

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